

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the quadratic equation $x^2 - 2(k + 1)x + k^2 = 0$ has real and equal roots, the value of k is:
(A) (A) $1/2$
(B) (B) -1
(C) (C) $-1/2$
(D) (D) 0
2. If the prime factorisation of a positive integer n is given by $n = 2^3 \times 3^2 \times 5^a$ and it is known that the decimal expansion of the rational number $\frac{7}{n}$ terminates after exactly two decimal places, then the value of a must be:
(A) (A) 1
(B) (B) 2
(C) (C) 3
(D) (D) 0
3. If the polynomial $p(x) = x^4 - 2x^3 + 3x^2 - ax + b$ is exactly divisible by the polynomial $g(x) = x^2 - 2x + 1$, then the values of a and b are respectively:
(A) (A) $2, 4$
(B) (B) $4, 1$
(C) (C) $4, 2$
(D) (D) $2, 2$
4. The pair of equations $\frac{4}{x} + 3y = 8$ and $\frac{6}{x} - 4y = -5$ is given. If these equations are transformed into a linear pair by substituting $u = \frac{1}{x}$, the value of u satisfying the system is:
(A) (A) $\frac{1}{2}$
(B) (B) 2
(C) (C) $\frac{1}{4}$

- (D) (D) 4
5. If the sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$, and the k^{th} term of this progression is 164, then the value of k is:
- (A) (A) 25
(B) (B) 26
(C) (C) 27
(D) (D) 28
6. The value of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal real roots is
- (A) (A) 8
(B) (B) 4
(C) (C) 2
(D) (D) 0
7. If one root of the quadratic equation $x^2 - 7x + m = 0$ is 4, then the other root is
- (A) (A) 7
(B) (B) 4
(C) (C) 3
(D) (D) 1
8. The quadratic equation $x^2 + 2x + 5 = 0$ has
- (A) (A) two distinct real roots
(B) (B) no real roots
(C) (C) two equal real roots
(D) (D) more than two real roots
9. The value of p for which the equation $px^2 - 2x + 1 = 0$ is not a quadratic equation is
- (A) (A) 2
(B) (B) 1
(C) (C) -1
(D) (D) 0
10. Which of the following is a quadratic equation?
- (A) (A) $(x + 1)^2 = 2(x - 3)$
(B) (B) $x^2 + 1/x^2 = 5$
(C) (C) $(x + 2)^3 = x^3 + 1$
(D) (D) $x(x + 1) = x^2 + 5$

11. If α and β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ is
- (A) (A) 25
 - (B) (B) 10
 - (C) (C) 13
 - (D) (D) 12
12. The roots of the equation $x^2 - 3x - 10 = 0$ are
- (A) (A) 2, -5
 - (B) (B) 5, -2
 - (C) (C) 5, 2
 - (D) (D) -5, -2
13. If the discriminant of $3x^2 - 2x + k = 0$ is 4, the value of k is
- (A) (A) 1
 - (B) (B) $1/3$
 - (C) (C) 2
 - (D) (D) 0
14. The sum of the roots of the equation $2x^2 - 8x + 6 = 0$ is
- (A) (A) 4
 - (B) (B) 8
 - (C) (C) -4
 - (D) (D) 3
15. Assertion (A): The quadratic equation $2x^2 + kx + 8 = 0$ has real and equal roots for $k = 8$ or $k = -8$. Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has real and equal roots if and only if its discriminant $D = b^2 - 4ac$ is equal to 0.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
16. Assertion (A): The number $\frac{7}{2^3 \times 5^2}$ has a terminating decimal expansion. Reason (R): For a rational number $\frac{p}{q}$ to have a terminating decimal expansion, the prime factorisation of the denominator q must be of the form $2^n \times 5^m$, where n and m are non-negative integers.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
17. Consider the following Assertion and Reason regarding the division algorithm for polynomials $p(x) = g(x)q(x) + r(x)$, where $p(x)$ is a polynomial of degree 4 and $g(x)$ is a polynomial of degree 2. \\ \\ Assertion (A): The degree of the quotient $q(x)$ must be exactly 2. \\ Reason (R): According to the division algorithm, the degree of the remainder $r(x)$ must be less than the degree of the divisor $g(x)$, and the degree of the product $g(x)q(x)$ must equal the degree of the dividend $p(x)$ if the remainder is zero or of lower degree.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
18. Assertion (A): The pair of equations $\frac{1}{x} + \frac{2}{y} = 4$ and $\frac{2}{x} + \frac{4}{y} = 12$ is inconsistent. Reason (R): A pair of linear equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is inconsistent if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

- Find the value of k for which the quadratic equation $(k-2)x^2 + 2(2k-3)x + (5k-6) = 0$ has real and equal roots.
- If the HCF of 120 and 225 is expressed in the form $225 \times 5 + 120 \times k$, find the value of k .
- If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $g(x) = x^2 - 2x + k$ and the quotient is $x^2 - 4x + 8$, determine the value of the constant k and the remainder $r(x)$.
- Find the value of k for which the pair of equations $\frac{1}{x} + \frac{2}{y} = 4$ and $\frac{2}{x} + \frac{k}{y} = 10$ is inconsistent, given that $x, y \neq 0$.

Section C: Short Answer (3 marks each)

1. Show that the square of any positive integer n is of the form $3m$ or $3m + 1$ for some integer m , and use this result to determine if the square of an integer can be of the form $3m + 2$.
2. Find the value of k for which the quadratic equation $(k + 1)x^2 - 2(k - 1)x + 1 = 0$ has two equal real roots. Further, find the roots of the equation for this value of k .
3. Show that for any positive integer n , the square of a positive integer cannot be of the form $5m + 2$ or $5m + 3$ for any integer m .
4. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $g(x) = x^2 - 2x + k$, the remainder is found to be of the form $x + a$. Find the values of k and a .
5. Solve the following pair of equations for x and y by reducing them to a linear form:
 $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, where $x \neq 0$ and $y \neq 0$.

Section D: Long Answer (4-5 marks each)

1. A train travels at a certain average speed for a distance of 630 km. If the speed had been 15 km/h more, the journey would have taken 2 hours less. Find the original average speed of the train.
2. Show that for any positive integer n , the expression $n^3 - n$ is always divisible by 6.
3. A rectangle has a length x and a width $x - 3$. If the area of this rectangle is numerically equal to the perimeter of a square with side x , find the quadratic equation representing this condition.
4. A train travels at a certain average speed for a distance of 540 km. If the speed of the train is increased by 9 km/hr, the journey would have taken 2 hours less. Find the original average speed of the train. Also, find the time taken if the train travels at the increased speed.
5. Show that for any positive integer n , the expression $n^3 - n$ is always divisible by 6. Use the Fundamental Theorem of Arithmetic or properties of divisibility to justify your answer.
6. If the polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $g(x) = x^2 - 2x + k$, the remainder is found to be of the form $x + a$. Find the values of k and a . Hence, determine all the zeros of the polynomial $p(x)$.
7. A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and that of the boat in still water.

Section E: Case Study (4 marks each)

1. A group of students is designing a rectangular garden for a school project. The length of the garden is planned to be 3 meters more than twice its width. They have a total of 20 square meters of space available for the garden. Due to the placement of a nearby path, the actual area must be exactly 20 square meters to maintain symmetry. Let the width of the garden be x meters.
 - (i) Which of the following quadratic equations correctly represents the area of the garden?
 - (A) (A) $2x^2 + 3x - 20 = 0$
 - (B) (B) $x^2 + 3x - 20 = 0$
 - (C) (C) $2x^2 + 3x - 20 = 0$
 - (D) (D) $2x^2 - 3x - 20 = 0$
 - (ii) Find the discriminant D of the quadratic equation obtained in the previous part.
 - (iii) What is the width of the garden in meters?
 - (A) (A) 2.5
 - (B) (B) 4
 - (C) (C) 3
 - (D) (D) 2
 - (iv) Calculate the length of the garden based on the width found.

2. A packaging company produces two types of luxury gift boxes, Type A and Type B. The number of chocolates in Type A boxes is 120 and in Type B boxes is 168. The quality control manager decides to pack these chocolates into smaller identical packets such that each packet contains the same number of chocolates, and no chocolates are left over. Additionally, the company wants to estimate the total number of packets required if the total supply of chocolates consists of 120 chocolates of Type A and 168 chocolates of Type B. Let H be the maximum number of chocolates that can be placed in each packet such that no chocolates are left over, and let L be the least common multiple of the number of chocolates in the two types of boxes.
 - (i) What is the maximum number of chocolates (H) that can be placed in each packet?
 - (A) (A) 12
 - (B) (B) 24
 - (C) (C) 36
 - (D) (D) 48
 - (ii) What is the value of L , the least common multiple of 120 and 168?
 - (A) (A) 420
 - (B) (B) 640
 - (C) (C) 840
 - (D) (D) 1020
 - (iii) If the company wants to pack all 120 chocolates of Type A into packets of size H , how many packets are needed?
 - (iv) Is the decimal expansion of $\frac{120}{168}$ terminating or non-terminating repeating?

3. A landscape architect is designing a rectangular meditation garden. The length of the garden is 3 m more than twice its width. A concrete path of uniform width 1 m is constructed around the inside of the garden, reducing the area available for planting to 80 m^2 . Let the width of the garden be x meters.
- Which of the following quadratic equations represents the area of the inner planting region?
 - (A) $2x^2 + x - 84 = 0$
 - (B) $2x^2 - x - 84 = 0$
 - (C) $2x^2 + 5x - 80 = 0$
 - (D) $2x^2 - 5x - 80 = 0$
 - Find the dimensions (width and length) of the entire rectangular garden.
 - What is the discriminant of the quadratic equation $2x^2 - 3x - 82 = 0$?
 - If the outer area of the garden is 119 m^2 , find the width x .
4. A packaging unit at a local factory produces two types of high-quality metallic components in batches. The number of components in Batch A and Batch B are 120 and 192 respectively. The manager wants to pack these components into identical boxes such that each box contains the same number of items, and no components are left over. Additionally, the decimal representation of the ratio of the number of items in the two batches, $\frac{120}{192}$, is to be analyzed to ensure the packaging process meets specific efficiency standards.
- What is the maximum number of components that can be packed in each box such that no components are left over?
 - (A) 12
 - (B) 16
 - (C) 24
 - (D) 48
 - Express the simplified form of the fraction $\frac{120}{192}$ as a rational number $\frac{p}{q}$ where p and q are coprime.
 - Without performing long division, determine the nature of the decimal expansion of $\frac{120}{192}$.
 - (A) Terminating
 - (B) Non-terminating repeating
 - (C) Non-terminating non-repeating
 - (D) None of these
 - If the total number of components 120 and 192 were to be used to form a new larger batch, calculate the LCM of the two numbers.
5. An engineer is designing a robotic path for a manufacturing unit. The trajectory of the robot is modeled by a polynomial $p(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$. Due to a sensor constraint, the path must be analyzed relative to a base movement polynomial $g(x) = x^2 - 2x + k$. When $p(x)$ is divided by $g(x)$, the resulting remainder is found to be of the form $ax + b$. Based on this division algorithm, answer the following questions:

- (i) If the division of $p(x)$ by $g(x)$ leaves a remainder $x + a$, what must be the value of k ?
- (A) (A) 3
 (B) (B) 4
 (C) (C) 5
 (D) (D) 2
- (ii) Using the value of k found in the previous step, find the value of the constant 'a' in the remainder $x + a$.
- (iii) What is the quotient obtained when $p(x)$ is divided by $x^2 - 2x + 5$?
- (A) (A) $x^2 - 4x + 3$
 (B) (B) $x^2 + 4x + 3$
 (C) (C) $x^2 - 4x - 3$
 (D) (D) $x^2 + 4x - 3$
- (iv) If the remainder of the division of $p(x)$ by $g(x)$ were 0, what would $g(x)$ be called with respect to $p(x)$?
6. A water treatment plant uses two types of filtration units, Type A and Type B. The efficiency of the plant is modelled by the following equations involving the flow rates x (in litres/sec) and y (in litres/sec) through these units. The plant operator observes that the system follows the relations: $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, where $x, y \neq 0$.
- (i) If we substitute $u = \frac{1}{x}$ and $v = \frac{1}{y}$, what is the resulting linear equation from the first condition?
- (A) (A) $2u + 3v = 13$
 (B) (B) $3u + 2v = 13$
 (C) (C) $2u + 3v = 1$
 (D) (D) $13u + 13v = 1$
- (ii) Find the values of u and v .
- (iii) What are the values of the flow rates x and y ?
- (A) (A) $x = 3, y = 2$
 (B) (B) $x = \frac{1}{2}, y = \frac{1}{3}$
 (C) (C) $x = 2, y = 3$
 (D) (D) $x = \frac{1}{3}, y = \frac{1}{2}$
- (iv) If the plant introduces a new constraint $\frac{1}{x} + \frac{1}{y} = k$ such that the system remains consistent with the previous solution, find the value of k .